

Explainability-Aware Adversarial Threats and Mitigation in Federated Learning Based Anomaly Detection for Cooperative Smart Farming

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Abstract—Cooperative Smart Farming (CSF) provides an effective solution to address the evolving needs of smart farming, making precision agriculture more accessible to small-scale farmers. These cooperatives are formal enterprises collectively financed, managed, and operated by member farms, working together for shared benefits. However, CSFs face increased security risks, since a cyberattack on one farm can disrupt the entire network, threatening data integrity and decision-making. Federated Learning (FL) offers a robust solution that enables distributed learning by maintaining a global model across the cloud server and multiple client farms on each edge node, where the global model is trained on the client's model parameter without accessing the client's private data.

This paper demonstrates that FL-based network anomaly detection systems in CSF are vulnerable to data poisoning adversarial attacks. We present a novel poisoning attack strategy in which an adversary identifies and exploits the most essential features of the dataset using Explainable AI techniques. Using the Explainability-informed features, the adversary can perform targeted data poisoning attacks using the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) attacks against the trained Convolutional Neural Network (CNN) classifier. Our experimental results indicate that the accuracy of the FL model declines significantly when adversaries poison data by manipulating essential features compared to random perturbations. We also propose a defense mechanism leveraging DistilBERT, a lightweight language model deployed on each local client to mitigate this adversarial attack. Our defense approach effectively filters out poisoned data using cosine similarity, restoring model robustness and accuracy.

Index Terms—Cooperative Smart Farming, Network Anomaly Detection System, Federated Learning, Explainability, Adversarial Attacks, Data poisoning Attacks, DistilBERT, Fast Gradient Sign Method

I. INTRODUCTION

In response to the increase in global population, farmers have started to utilize advanced technologies like the Internet of Things (IoT), Machine Learning (ML), cloud computing, and autonomous systems to effectively manage the balance between food supply and demand [1], [2]. Although more farmers are eager to adopt data-driven technologies, the cost and technical expertise required often present significant obstacles for small/medium-scale farmers. A recently proposed approach by Praharaj et al. [3], [4] presented Cooperative Smart Farming (CSF) which benefits the small/medium-scale member farms by providing facilities such as resource sharing,

financial assistance, access to large machinery, efficient labor during peak fieldwork periods, production transportation, price negotiation, equipment repair and maintenance, and data sharing, among others [5], [6]. However, as per the recent research, and a report by the Federal Bureau of Investigation (FBI), cyberattacks on agricultural systems are no longer a distant threat but a growing reality [7], [8]. In CSF, however, the risks are more intense, as an attack on a single farm can compromise all member farms due to their interconnected networks and extensive data sharing [3], [4].

To mitigate this issue, farmers can deploy Network Anomaly Detection Systems (ADS) in each smart farm within the CSF network to detect and identify compromised farms. An ADS passively monitors and analyzes device communications (i.e. network traffic) to detect suspicious activities and determine if the network is under attack [3]. An ADS based solution has the potential to detect unknown attacks at its local network but fails to detect any unseen (i.e. zero-day) cyberattacks on other farms within CSF [4]. In this context, Federated Learning (FL) is an appropriate fit, as it provides for the distributed training of ML models that share the anomaly profiles with the other member smart farms and take preventive measures [4], [9]. In FL, each local smart farming client participating in the system uses its private network dataset to train a local AD model and send the model aggregates to a central aggregator. The aggregator then uses a federated averaging algorithm to aggregate the local models into a global model, which is then distributed back to the member smart farm clients [4], [9].

Despite the efficiency of FL in network ADS, researchers often overlook its vulnerability to adversarial attacks. FL introduces security concerns as its clients are susceptible to even minor changes in input data [10], [11]. A slight manipulation or introduction of perturbations can lead to unintended outputs in the global FL model [12]. For instance, an attacker could perturb the training data of a malicious client to forcibly gain access to a resource or poison the data by crafting adversarial samples. This could compromise the global FL accuracy by misclassifying malicious network attacks as benign or vice versa [13].

Explainable AI (XAI) methods have been introduced to enhance FL model transparency and security by focusing on feature attribution, clarifying model decisions, and im-

proving user trust. These explanations play a crucial role in determining important features for network ADS, empowering security analysts in countering threats [14], [15]. However, explanations can also aid in generating more effective adversarial samples for data poisoning attacks. By revealing crucial features in model decision-making, an adversary can craft samples that intentionally mislead the FL classifier, causing misclassification and evade anomaly detection. Several methods [16] have been proposed to mitigate adversarial attacks, including Krum, Multi-Krum, Trimmed Mean, Bulyan, and Fed-Def. However, these strategies primarily address model poisoning attacks, leaving data poisoning attacks at the client level less explored [16].

This paper employs Shapley Additive Explanations (SHAP) to interpret model decisions and inform targeted misclassification attacks on the FL model. A CNN-based binary classifier is utilized, with DeepSHAP providing global insights into model behavior. These explanations guide adversarial attacks in binary classification tasks, leverage the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) for adversarial analysis. Further, to make model robust against these data poisoning attacks, we propose a comprehensive defense strategy that effectively eliminates adversarially poisoned samples at the client level, ensuring robust FL training. As part of our approach, we deploy a lightweight LLM model, DistilBERT, at each FL client farm. Each client extracts embeddings from its training samples using the deployed model. Then, the cosine similarity between each sample and a clean reference dataset is computed. The sample is flagged as poisoned and removed if the similarity falls below a predefined threshold, resulting in a denoised dataset. The FL client subsequently trains on the denoised dataset, improving model robustness. Evaluation results show that this approach effectively mitigates data poisoning attacks, leading to a significant accuracy improvement in the FL model post-attack. The main contributions of the papers are as follows:

- 1) We evaluate the effectiveness of an FL-based CNN model for network anomaly detection using Smart Farming datasets.
- 2) To enhance the analysis, we employ Shapley Additive Explanations (SHAP) to interpret model predictions. These SHAP explanations are then leveraged to inform targeted data poisoning attacks, leading to misclassification in the Global FL model.
- 3) We evaluate adversarial attacks using two feature selection strategies: the first four features and randomly selected four features. These evaluations are conducted under two attack scenarios: a single-farm attack and a multiple-farm attack, assessing their impact on the FL model's performance.
- 4) As part of our adversarial defense strategy, we deploy a DistilBERT model at each FL client and filter out poisoned data using cosine similarity-based masking.
- 5) Finally, we re-evaluate model performance on the denoised dataset and compute the FL accuracy, assessing

the model's robustness against data poisoning attacks.

The remainder of this paper is organized as follows. Section II discusses the relevant literature. Section III addresses the proposed adversarial attack method. Section IV defines the dataset and the attack setting. Section V explains the results of the adversarial attack. Section VI provides the defense strategy for mitigating the performed adversarial attack. The conclusion and future works are presented in Section VII.

II. RELATED WORK

A. Adversarial attack and Defense on Federated Learning

FL is a distributed ML approach that enhances the privacy of training datasets while optimizing communication efficiency during training rounds. Due to these advantages, FL has been extensively adopted in network ADS to improve security and threat detection without compromising sensitive data [4], [9], [17], [18]. However, recent research indicates that FL is vulnerable to poisoning attacks [13], [19], [20], where an adversary compromises the global model by manipulating the client's training data or model updates, leading to degraded model performance and biased decision-making. Biggio et al. [21] introduced the first poisoning attack method against support vector machines (SVM). The authors used a gradient ascent strategy to predict changes in the SVM's decision function, thereby degrading the target model's accuracy. Subsequently, Muñoz-González et al. [22] extended poisoning attacks to deep learning models by leveraging the back-gradient optimization method, enabling these attacks to target a broader range of deep learning algorithms. Bagdasaryan et al. [20] proposed the first backdoor attack model against FL, where one or more attackers embed backdoor functionality into the FL using a generic constrain-and-scale technique. Later, Fang et al. [23] explored the first local model poisoning attacks targeting Byzantine-robust FL methods. These attack strategies primarily aim to alter the learning trajectory of the model, causing deviations and ultimately shifting the decision boundaries of the global model. In [13], the authors presented a novel data poisoning attack that allows an adversary to implant a backdoor into the aggregated detection model to incorrectly classify malicious traffic as benign.

As the diversity and complexity of adversarial attacks against FL increase, several defense mechanisms are being developed to counteract them. These defense mechanisms are broadly categorized into Data Sanitization, Data Anomaly Detection, Adversarial training, model compression, and Byzantine-resilient aggregation [10]. Adversarial training, which involves augmenting the training dataset with adversarial examples, is one of the most straightforward and widely used defense strategies. By training the model on both clean and adversarially perturbed inputs, it learns to identify and resist such perturbations. However, the existing models may not generalize well to new attacks not seen during training [24]. Our work aims to build upon the fundamentals of adversarial training, exploring new strategies to enhance the robustness of FL-based network anomaly classifiers against

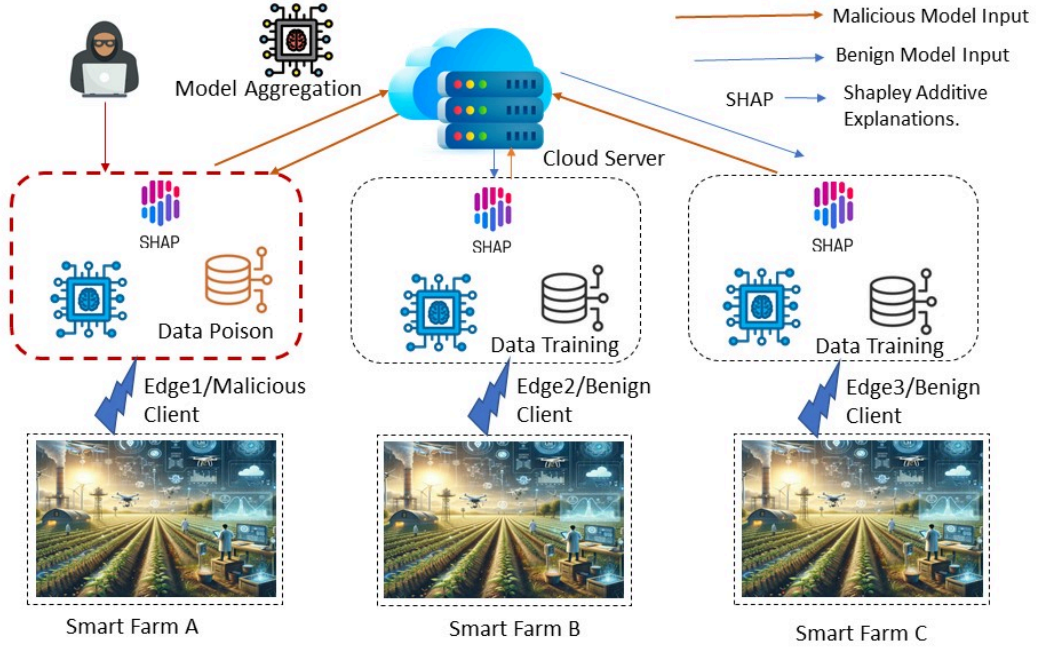


Fig. 1: Overview Of the Adversarial Attack on FL-based Network Anomaly Detection System

FGSM and PGD attacks, thereby contributing to developing more secure and reliable FL systems.

III. PROPOSED METHOD

A. Overview of the Attack

This work considers FL-based for cooperative smart farming on Network Anomaly Detection tasks (Fig. 1). The setup consists of one server S , M local ADS clients $K = \{k_1, k_2, \dots, k_M\}$, and M private training datasets:

$$D_{\text{train}} = \{D_{\text{train}_1}, D_{\text{train}_2}, \dots, D_{\text{train}_M}\}$$

We assume that from the total client farms M in CSF, the attacker can control N of them by injecting adversarial perturbations into the local training process ((Fig. 1)). The client farms not controlled by the attacker are called benign clients, and those controlled by the attacker are called malicious clients. The attackers aim to reduce the accuracy of the global FL model by whitebox attacks, which require knowledge of the target model architecture. The objective is to misclassify the benign packet in the training network dataset as malicious and vice versa.

Due to the nature of FL, we assume that the attacker can arbitrarily modify the malicious client's local training process, for example, modifying the training inputs or even updating the local updates without affecting the benign clients' local update and server aggregation algorithm. The attacker also

knows the model information of the malicious client, the features of the data set, and the server aggregation algorithm. However, the attacker does not know about the updates on the benign client (Fig. 1). The attacker's strategy to perform a data poisoning attack on the FL-based Network ADS is as follows:

- 1) The server S first sends the global parameter to all the clients.
- 2) The benign client downloads the global parameter, performs the local training on private network data and sends the model update to the server.
- 3) The malicious client downloads the global parameter, performs the local training on the adversarial samples and then sends the malicious update to the server S .
- 4) Server S performs the model aggregation and repeats steps 1) to 3) until the model converges.

B. Attack Methods

For targeted data poisoning attacks, we choose the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) attacks [25], both white box attacks requiring the knowledge of the client's model architecture. We aim to generate adversarial samples that are misclassified as benign or malicious to assess the robustness of the global FL-based ADS. The FGSM method is a fast yet effective method that produces adversarial samples with one-step calculation. Assuming input x and its corresponding target t , FGSM calculates the gradient of x with respect to the loss function. The adversarial example

is generated using the following formula:

$$x = x + \epsilon \cdot \text{sgn}(\nabla_x L(g(x; w), t)) \quad (1)$$

where ϵ is a hyper-parameter that determines the adversarial noise level, $g(x; w)$ is the output of the neural network with respect to the input x and parameter set w , and $\text{sgn}(\cdot)$ is the sign function.

The sign function ensures that only the direction of the gradient is used to construct the perturbation, scaled by ϵ to control its magnitude:

$$\epsilon \cdot \text{sgn}(\nabla_x L(g(x; w))) \in [-1, 1]. \quad (2)$$

In PGD, the attack starts with a uniform random initialization. The update formula for the PGD attack is given by:

$$\hat{x}^{t+1} = \Pi_{P_\epsilon(x)}(\hat{x}^t + \alpha \cdot \text{sgn}(\nabla_x L(g(\hat{x}^t; w), t))) \quad (3)$$

where, $\hat{x}^t$ is the perturbed data at the t -th iteration, $P_\epsilon(x)$ is the projected gradient descent function, $\text{sgn}(\cdot)$ is the sign function, $L(g(\hat{x}^t; w), t)$ is the loss function with respect to the perturbed input $\hat{x}^t$ and the neural network parameters w , and α is the step size for each iteration. The projection function $P_\epsilon(x)$ ensures that the perturbations remain within a small ϵ -neighborhood of the input x . This is achieved by first calculating the sign values of the gradient and then projecting the results back to the allowable perturbation space. PGD is considered one of the strongest adversarial attacks and is a universal first-order adversarial method.

To inform the attacks, we first utilized SHAP (Shapley Additive Explanations) [26] to identify the features most critical to the local model's decision-making process (Algorithm 1). The SHAP values are a common way of getting consistent and objective explanations of how each feature impacts the model's prediction. Unlike related works that focused on targeting transparent models, our approach leverages post-hoc explanations to pinpoint the most influential features in the network traffic dataset (Fig. 2). This targeted identification reduces the attack vector, making the data poisoning attacks more precise and effective in FL-based Network ADS (Algorithm 1, (Fig. 2)).

Algorithm 1 (Algorithm 1, (Fig. 2)) illustrates the adversarial attack strategy in an FL-based ADS for CSF. This approach establishes an FL framework, enabling multiple clients to train an anomaly detection model collaboratively without sharing raw data. The attacker then identifies the most critical features for adversarial perturbation using SHAP values. Subsequently, adversarial samples are generated using the FGSM and PGD methods. Each malicious client evaluates the effectiveness of these adversarial samples on its local anomaly detection classifier. Finally, the server aggregates and integrates the adversarially perturbed data to assess its overall impact on the global model.

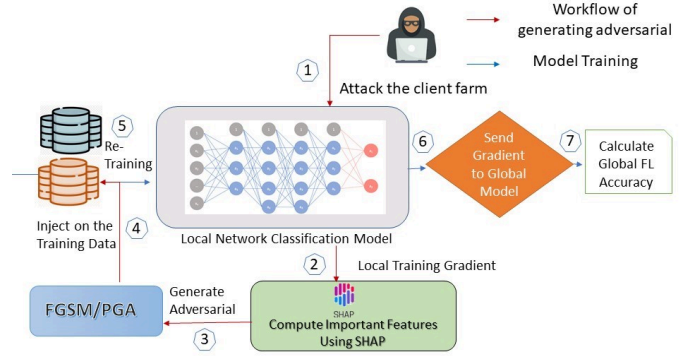


Fig. 2: Overview of the Proposed Data Poisoning Attack

Algorithm 1: Explainability Guided Adversarial Attack in FL based ADS

Data: A network anomaly detection classifier, network data $\mathbf{D}_k$ of all the malicious clients, size of perturbation, number of features to perturb, SHAP values

Result: Accuracy of the trained global model

1. Initialize Federated Learning Environment for Network Anomaly Detection;

2. Configure attack parameters for targeted misclassification in the FL context;

for each malicious client do

3. Identify the most important features using SHAP values;

4. Use FGSM and PGD to generate adversarial examples by perturbing the identified important features;

5. Evaluate the local model on the adversarial examples to measure the accuracy locally;

for Server-Side Aggregation do

6. Aggregate adversarial example results from clients to simulate their impact on the global model;

7. Evaluate the global FL classifier on the generated adversarial examples;

8. Calculate the accuracy for FGSM and PGD attacks on FL;

IV. EXPERIMENTAL DESIGN

A. Dataset and Model Architecture

In this research, we used two raw datasets from our past research [3], [4], which were classified into two different smart farms, Smart Farm 1 and Smart Farm 2 datasets. We preprocess this data set by removing any missing or malformed values. Standardization was applied to all captured features by calculating their standard deviations to ensure consistency across the data. Additionally, we removed certain unnecessary features for the network anomaly detection system. Features like 'Frame number', 'Source IP', 'Destination IP', 'Timestamp', 'Source port', and 'Destination port' were excluded to



Fig. 3: SHAP analysis of Smart Farm 1 Dataset

prevent the models from simply learning patterns based on numbering on specific IP addresses and ports, which could lead to overfitting or biased predictions. We performed one-hot encoding to transform the categorical or string values into integers. The data was then split into training and test sets at a ratio of 3:1. Again, the training dataset was split into training and validation sets at 3:1.

Our global model uses a neural network model based on CNN architecture. The model leverages a CNN to perform binary classification on network data. It is designed to extract hierarchical feature representations from the input while preserving computational efficiency. The architecture begins with two convolutional layers interspersed with max-pooling operations to reduce the temporal dimensions and capture meaningful local patterns progressively. The output of the convolutional layers is flattened into a 1D vector and passed through a fully connected dense layer with 128 neurons and ReLU activation, which combines the extracted features into a compact representation. To prevent overfitting, a dropout layer with a rate of 50% is incorporated. The final output layer consists of a single neuron with a sigmoid activation function, producing a probability score for binary classification. The model is optimized using the Adam optimizer, with binary cross-entropy as the loss function.

B. Global Explanation

After setting up the global FL model, we compute the SHAP values to inform the attacks using SHAP's Deep Explainer. We used a subsample of smart farm clients' training data

from the training dataset. This under-sampling is required due to the resource-intensive nature of computing SHAP values balanced with the need to create effective adversarial samples. The summary plots in Fig. 3 illustrate essential features across the Smart Farm client's dataset. The X-axis shows the average magnitude impact on the model output, with the features arranged by their effect's magnitude. The key features identified in the model trained on Smart Farm1's client dataset were predominantly TCP, ICMP and HTTP attributes, highlighting their significance in network traffic analysis. We used the top four identified features for each respective client farm's dataset to inform the target data poisoning attack.

C. Simulation setup and Attack Settings

In this work, the Flower framework [27], an FL-based platform, was utilized to implement the proposed algorithm. Flower supports the Python programming language and enables large-scale FL simulations. It supports various distributed parameter aggregation algorithms like FedAvg, FedAvgM, FedAdam, FedBN, and others while seamlessly integrating with TensorFlow, PyTorch, and scikit-learn. The proposed FL algorithm was executed on an Intel Core i9 processor with 64 GB of memory, which acted as the central server for client connections and model aggregation. Two additional Intel Core i9 processors with 32 GB of memory were used on the client side. Flower was installed on these machines to facilitate communication and integration between the server and clients.

The training configuration included a learning rate of 0.001 using the Adam optimizer and a batch size of 32. EarlyStopping and ReduceLROnPlateau were employed to optimize and speed up the training process of neural network models. In our research, the attacker uses algorithm 1 to generate adversarial samples by adding perturbation and accumulating the velocity vector of the loss function during iteration (Fig. 2). We performed a random search to find out the optimal hyperparameter for our misclassification task. For FGSM, the epsilon (ϵ) is 0.001, and for PGD the epsilon is 0.0001 with alpha (α) 0.005. We incorporated adversarial samples into 30% of the training data for each attack. The attacker optimizes FL learning using the FedAvg algorithm 1 in our experiments.

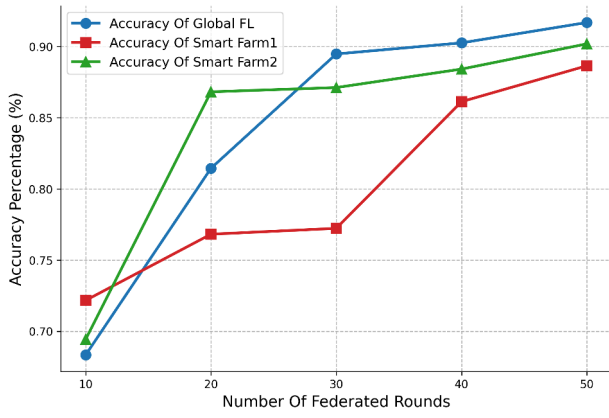
D. Evaluation Matrices

$G_{adv.acc}$ denotes the accuracy of the global model when exposed to an adversarial attack, whereas $G_{def.acc}$ represents the accuracy of the global model after applying adversarial defense. The improvement in accuracy due to the defense mechanism is represented by δ , which is computed as: $\delta = G_{def.acc} - G_{adv.acc}$.

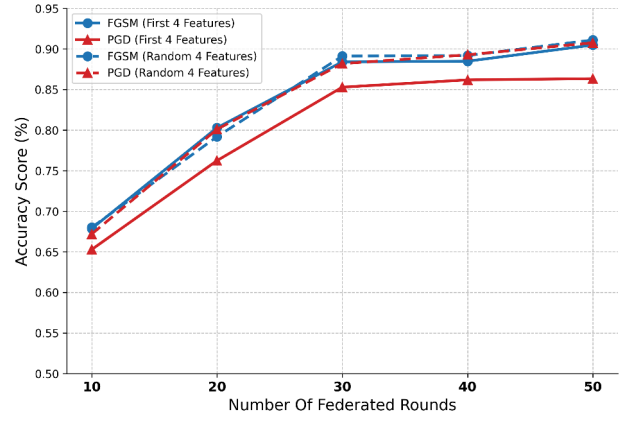
V. RESULT AND DISCUSSION

A. Impact of Adversarial Attacks on Global Federated Learning Model

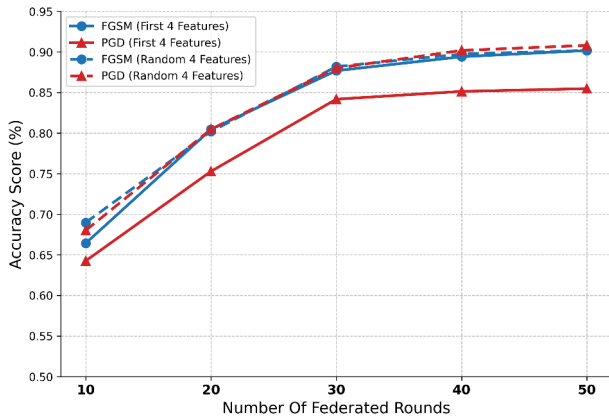
Figure 4 compares the global Federated Learning (FL) model accuracy under normal and adversarial attack scenarios. Figure 4a illustrates the global model accuracy without an adversarial attack. Figure 4b and Figure 4c depict the impact of adversarial attacks on Smart Farm 1 and Smart Farm 2, respectively. Figure 4d demonstrates the global model



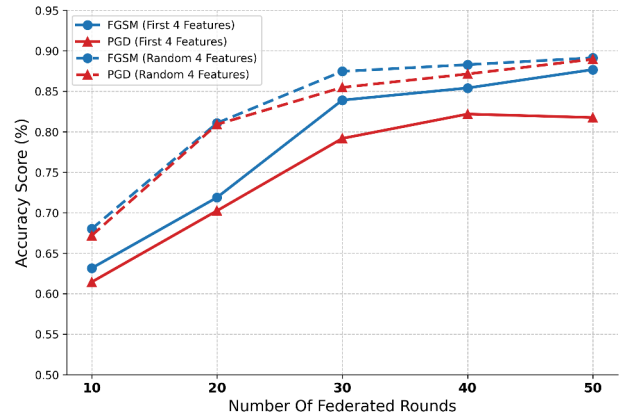
(a) Performance analysis of FedAVG Algorithm without attack



(b) Performance analysis of FedAVG Algorithm when Smart Farm1 is under attack



(c) Performance analysis of FedAVG Algorithm when Smart Farm2 is under attack



(d) Performance analysis of FedAVG Algorithm when both the Smart Farms are under attack

Fig. 4: Performance Analysis of FedAvg Algorithm with Number of Federated Rounds Under Different Attack Scenario.

accuracy when Smart Farm 1 and Smart Farm 2 are attacked. To perform the attack, We first extract four important features using the DeepSHAP algorithm and then induce poisoned data using FGSM and PGD attack methods. Next, we compute the global FL accuracy using the FedAVG algorithm. We repeat the same process with four randomly selected features from the training samples.

As shown in Figure 4b, the global model accuracy of FL decreases under attack, and the training curves become flatter when Smart Farm 1 is compromised. From the figure, we can observe that the FGSM method is more effective when applied to the first four important features than a random selection of four features. The global FL accuracy decreases by 1% when using the critical features of the training data. However, the PGD attack is more effective, reducing the global FL accuracy from 91% to 86% when targeting these important features in Smart Farm 1 training data. This indicates that adversarial attacks are more successful when the adversary targets the essential features of the training data. The results remain consistent when poisoning the training data of Smart

Farm 2 using the same principles discussed above (Figure 4c). The global FL accuracy is not significantly affected when random features are used. However, in the case of Smart Farm 2, the PGD attack targeting the essential features proves to be more effective. As shown in Figure 4c, the global FL accuracy decreases from 91% to 85%, further reinforcing that adversarial attacks are more successful when key features of the training data are targeted.

B. Effects of Global FL accuracy on all malicious clients

In the next set of experiments, we poisoned the training data for both Smart Farm 1 and Smart Farm 2. Figure 4d illustrates the impact of malicious clients on FL training for FGSM and PGD attack methods. The results indicate that the effect on the global model intensifies as the percentage of malicious clients increases. For FGSM, when targeting the four important features, the global FL accuracy decreases from 90% to 87%, showing a more significant effect compared to a single-farm attack. The PGD attack proves even more effective, reducing the global FL accuracy to 81%. These findings suggest that the global FL accuracy is influenced by the number of malicious

clients and the selection of essential features in the training data.

VI. PROPOSED DEFENSE METHOD

Algorithm 2: Adversarial defense in Federated Learning based ADS using DistilBERT Model

Data: A global FL model for network anomaly detection, local datasets $\mathbf{D}_k$ from all clients, SHAP values, LLM-based filtering threshold

Result: Accuracy of the trained global model after adversarial defense

1. Initialize a Federated Learning environment for Network Anomaly Detection;
- for each FL client do**
 2. Deploy a lightweight LLM-based model at each client;
 3. Identify and filter poisoned data using LLM-based embedding similarity;
 4. Remove adversarial examples that deviate from the clean data threshold;
 5. Train a local CNN classifier on the denoised dataset;
 6. Send model updates to the FL server;
- for Server-Side Aggregation do**
 8. Aggregate client models using the FedAvg strategy;
 9. Evaluate the global model on clean test data;
 10. Compare accuracy before and after adversarial filtering;

This section presents our Adversarial Defense Strategy to mitigate data poisoning attacks on FL-based ADS. Unlike the proposed defense method in section II, our defense strategy detects and filters out adversarially manipulated samples at the client level using cosine similarity on LLM-generated embeddings, ensuring that only clean data contributes to federated learning updates. This proactive defense mechanism enhances robustness against poisoning attacks, reducing their impact on global model performance while maintaining privacy in federated settings.

The algorithm shows a detailed explanation of our proposed approach. First, a lightweight large language model (LLM) (DistilBERT) is deployed for each client (Algorithm 2). Previous work on continuous adversarial training on token embeddings has mostly focused on encoder-decoder models, such as BERT [28]. We opted for DistilBert as it is lightweight and faster. The pre-trained LLM (DistilBERT) converts each data sample into a vector representation called embedding. The embedding is a numerical vector that captures the semantic meaning of the input. For a data sample x , its embedding is represented as:

$$E_x = \text{LLM}(x)$$

Then, we compute the cosine similarity between the embedding of the new sample and the embeddings of known clean data. The cosine similarity between the embedding of the new sample E_x and the embedding of a clean reference sample E_c is given by:

$$\cos(\theta) = \frac{E_x \cdot E_c}{\|E_x\| \|E_c\|}$$

where:

- $E_x \cdot E_c$ is the dot product of the two embedding vectors.
- $\|E_x\|$ and $\|E_c\|$ are the Euclidean norms (magnitudes) of the vectors.
- $\cos(\theta)$ represents the similarity score between the two embeddings.

If the cosine similarity falls below a defined threshold, the sample is likely poisoned and is subsequently removed, resulting in a denoised dataset. The client then trains its model using this denoised dataset and submits the updated model to the Global FL model for aggregation. Each client undergoes a verification process to identify and eliminate poisoned data utilizing this strategy. The model gradients are computed and sent to the Global FL model, where the overall accuracy is evaluated (Algorithm 2).

A. Evaluation

Feature	After Attack ($G_{adv,acc}$)	After Defense ($G_{def,acc}$)	δ
FGSM (Smart Farm 1)	90.48	91.16	0.68
PGD (Smart Farm 1)	86.34	89.04	2.7
FGSM (Smart Farm 2)	90.18	91.49	1.31
PGD (Smart Farm 2)	85.48	89.04	3.56
FGSM (Both Smart Farm)	87.68	90.16	2.48
PGD (Both Smart Farm)	81.76	89.34	7.58

TABLE I: Comparison of our defense Method in Smart Farm Datasets

We evaluated our defense strategy using Algorithm 2 and presented the results in Table I. For this evaluation, we focused on the first four features under FGSM and PGD attacks, as they have been identified as the most effective attack methods, as discussed in Section V.

To implement our defense strategy, we leverage the DistilBERT model, utilizing the "transformers" and "sentence-transformers" libraries to enhance adversarial robustness. We first generate text embeddings from the input data and computed their cosine similarity scores. We set a threshold of 0.7 for cosine similarity, which was determined through extensive calculations and averaging across multiple experiments. Data samples falling below this threshold were identified as potentially poisoned and masked from the training set. Following this, we retrained the model using the denoised dataset and aggregated the updated model at the FL server.

As shown in Table I, our defense strategy significantly improved the accuracy of the global model by mitigating the effects of adversarial attacks. For instance, in the case of the PGD attack, when both Smart Farm clients were poisoned, the global model's accuracy dropped to 81.76%. However, after applying our defense strategy, the accuracy increased to 89.34%, demonstrating a substantial recovery.

These results highlight the effectiveness of our approach in defending against targeted data poisoning attacks, ensuring improved robustness and reliability of the FL model.

VII. CONCLUSION AND FUTURE WORK

In this work, we explore the impact of adversarial samples on a FL-based Anomaly Detection System for CSF. We propose a poisoning attack method leveraging Explainable AI, where adversarial samples are generated by targeting important dataset features. Our results indicate that when an attacker manipulates the key features of the dataset, the attack is more effective compared to perturbing random features. We conducted poisoning attacks in two scenarios: Poisoning individual Smart Farm clients separately and attacking both Smart Farm clients simultaneously. Our findings reveal that as the number of malicious clients increases, the impact of the attack intensifies, leading to a more severe degradation in model performance. To counteract this, we implemented a defense strategy using DistilBERT. By computing text embeddings and applying cosine similarity filtering, we identified and removed poisoned data using a carefully chosen threshold. Our results demonstrated that this approach effectively mitigates adversarial attacks, restoring the FL model's accuracy. In future, we aim to expand our experiments by including more Smart Farm clients to analyze the broader impact of adversarial samples. We can also explore other LLM models for adversarial defense and conduct a comparative analysis of different defense strategies to enhance the robustness of FL-based ADS in Cooperative Smart Farming.

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